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ROYAL UNIVERSITY OF PNHOM PENH

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Design and Develop a Simple Learning Management System

A Research Report

In Partial Fulfilment of the Requirement for the Degree of
Bachelor of Information Technology Engineering

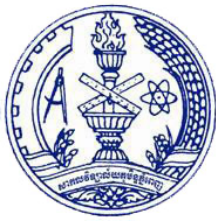
លីម ប៊ុនហួវ

LIM BUNHOV

លាង ច័ន្ទម៉ូលីកា

LEANG CHANMOLIKA

June 2025



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June 2024

មូលន័យសង្ខេប

ការអប់រំតាមប្រព័ន្ធអនឡាញក្លាយជាជម្រើសសំខាន់សម្រាប់ស្ថាប័នអប់រំជាច្រើនក្នុងការបន្តផ្តល់ការសិក្សាដល់សិស្សនិស្សិតក្នុងរយៈពេលដែលពិភពលោកប្រឈមមុខនឹងការរីករាលដាលនៃជំងឺកូវីដ-១៩។ ការបង្ហាញខ្លួនរបស់ជំងឺនេះបានបណ្តាលឲ្យសាលារៀន និងសាកលវិទ្យាល័យភាគច្រើន ត្រូវផ្អាកការបង្រៀនផ្ទាល់ខ្លួនជាបណ្តោះអាសន្ន ដើម្បីការពារសុវត្ថិភាពសិស្សនិស្សិត និងគ្រូបង្រៀន។ ក្នុងបរិបទនោះ ការបង្រៀនតាមអនឡាញត្រូវបានអនុវត្តយ៉ាងទូលំទូលាយ ដើម្បីធានាថាការសិក្សាមិនរង ការរាំងស្ទះ។ សិស្សនិស្សិតអាចចូលរួមថ្នាក់រៀនតាមរយៈបច្ចេកវិទ្យាបច្ចុប្បន្ន ដូចជា Zoom, Google Meet, Microsoft Teams និងកម្មវិធីផ្សេងៗទៀត។ វាជួយឲ្យការបង្រៀនបន្តដោយរលូន ទោះបីជាអវត្តមានផ្ទាល់ក៏ដោយ។ បន្ទាប់ពីស្ថានភាពជំងឺបានធ្លាក់ចុះ ស្ថាប័នអប់រំមួយចំនួននៅតែបន្តប្រើវិធីសាស្ត្របង្រៀនតាមអនឡាញជាមួយនឹងការបង្រៀនផ្ទាល់ដោយបង្កើតប្រព័ន្ធសិក្សាប្រកបដោយភាពបត់បែន។ មិនត្រឹមតែសិស្សនិស្សិតនៅតែមានឱកាសបន្តការសិក្សានោះទេ ប៉ុន្តែសិស្សនៅថ្នាក់វិទ្យាល័យមួយចំនួន ក៏បានចូលរួមក្នុងការសិក្សាអនឡាញផងដែរ។ ការប្រកួតប្រជែងផ្នែកបច្ចេកវិទ្យា ក៏បានបង្កើតឱកាសសិក្សាផ្សេងៗបន្ថែមដូចជា ការចូលរួមក្នុងវគ្គបណ្តុះបណ្តាលអនឡាញនានា និងការប្រកួតចំណេះដឹងតាមបណ្តាញ។

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ABSTRACT

Online education has become an important alternative for many educational institutions to continue delivering lessons to students during the global outbreak of COVID-19. The emergence of this pandemic forced most schools and universities to temporarily suspend in-person classes in order to protect the health and safety of students and teachers. In response to this situation, online learning was widely adopted to ensure that education could continue without disruption. Students were able to attend classes through modern technologies such as Zoom, Google Meet, Microsoft Teams, and other platforms. These tools allowed the teaching and learning process to carry on smoothly, even in the absence of physical interaction. Even after the pandemic situation improved, some educational institutions continued to implement online learning methods alongside in-person instruction, creating a more flexible approach known as blended learning. This not only allowed university students to keep up with their studies, but also enabled high school students in certain areas to access online education. The rise of digital competition also created additional learning opportunities, such as participating in various online training courses and virtual academic contests.

In conclusion, online education has proven to be a crucial and effective method in meeting the demands of learning during emergency situations like a health crisis. It continues to lead education into a new digital era, highlighting the advancement of technology and the evolution of more open and adaptive learning approaches.

SUPERVISOR'S RESEARCH SUPERVISION STATEMENT

TO WHOM IT MAY CONCERN

Name of program: Bachelor of Information Technology Engineering

Name of candidate: **LIM Bunhov, LEANG Chanmolika.**

Title of research report: **Design and Develop a Simple Learning Management System**

This is to certify that the research carried out for the above titled master's research report was completed by the above-named candidate under my direct supervision. This report material has not been used for any other degree. I played the following part in the preparation of this research report:

Supervisor's name: **CHHIM Bunchhun**

Supervisor's signature:

Date.....

CANDIDATE’S STATEMENT

TO WHOM IT MAY CONCERN

This is to certify that the research report we, LIM Bunhov and LEANG Chanmolika, hereby present, entitled “**Design and Develop a Simple Learning Management System**” for the degree of Bachelor of Information Technology Engineering at the Royal University of Phnom Penh, is entirely our own work and, furthermore, that it has not been used to fulfill the requirements of any other qualification, in whole or in part, at this or any other university or equivalent institution.

No reference to, or quotation from, this document may be made without the written approval of the author.

Signed by LIM bunhov:

Signed by LEANG Chanmolika:

Date:

Sign by CHHIM Bunchhun:

Supervisor’s signature:

Date.....

ACKNOWLEDGMENTS

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Sincerely,

LIM Bunhov

LEANG Chanmolika

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CHAPTER 1

INTRODUCTION

1.1 Background to the study

In recent years, the rapid advancement of educational technology has reshaped the landscape of teaching and learning across the globe. From virtual classrooms to mobile-based learning tools, digital transformation in education has created new opportunities for flexibility, access, and innovation. However, the integration of technology into mainstream education remains uneven especially in developing countries where infrastructure limitations, high costs of proprietary systems, and lack of digital literacy still pose significant barriers. Traditional, face-to-face classroom teaching methods continue to dominate in many universities and training centers, including those in Cambodia. These conventional models, while valuable, often suffer from several shortcomings. Firstly, they offer limited flexibility in terms of time and location, making it difficult for learners with part-time jobs, family responsibilities, or remote living situations to access quality education. Secondly, such methods tend to rely on passive learning approaches, where students are expected to absorb information without much interaction, engagement, or adaptation to individual learning styles. This one-size-fits-all strategy fails to support diverse learners with different paces, needs, and preferences.

In contrast, online learning platforms have emerged as promising alternatives. Tools like Moodle, Google Classroom, and Microsoft Teams allow students to learn at their own pace, revisit learning materials, and receive continuous updates. Despite these benefits, many existing Learning Management Systems (LMS) still fall short in practice. Their interfaces can be overly complex, the systems are often costly to implement and maintain, and many lack important features that support user engagement and progress tracking. For example, instructors may find it difficult to track individual student performance or identify those at risk of falling behind. Students, on the other hand, may feel isolated without timely feedback, peer interaction, or access to structured content.

Another key limitation is the lack of real-time performance analytics. Many current systems do not offer instructors actionable insights into how well students are performing on a week-to-week basis. Without such data, educators cannot easily adapt their teaching strategies or provide targeted support.

Similarly, students may be unaware of their learning gaps until it's too late, leading to decreased motivation and lower course completion rates. These issues became especially evident during the COVID-19 pandemic, which forced educational institutions worldwide to rapidly adopt digital learning models. In Cambodia, the transition highlighted the urgent need for accessible, affordable, and easy-to-use platforms that could support blended or fully online learning. However, many institutions lacked the technical resources or budgets to adopt enterprise-level LMS solutions. By focusing on simplicity, cost-effectiveness, and essential functionality, the proposed LMS seeks to close the gap between traditional education and digital learning. It will offer a centralized, user-friendly platform where instructors can manage content, monitor student performance, and communicate effectively, while students can access learning materials, track their progress, and engage more actively with the course content.

1.2 Problem Statement

In today's digital age, university students still face significant barriers when trying to access and engage with educational resources online. One major challenge is the lack of an efficient, centralized Learning Management System that suits their specific needs. Many existing platforms are either too complex, too costly, or not user-friendly for students and instructors in smaller academic environments.

Students often hesitate to ask questions during lectures. Especially those seated at the back of the classroom because they worry about interrupting others or appearing unprepared. This lack of interaction makes it harder for them to fully understand the material. As a result, they seek additional learning resources outside the classroom, but struggle to find qualified instructors or structured materials for effective self-study.

Additionally, instructors are forced to rely on multiple disconnected tools like messaging apps, emails, or physical handouts to share information. This leads to inefficient communication, lost updates, and poor organization of educational content. Without a single platform to manage all learning activities, students have difficulty tracking their own progress, and instructors find it hard to monitor performance or identify learners who need support.

This project aims to solve these challenges by developing a practical and user-friendly LMS. The system will offer streamlined course management, interactive features, communication tools, and progress tracking to improve both teaching and learning experiences. It will be especially designed to fit the needs of university students and educators who lack access to costly enterprise solutions.

1.3 Aim and Objectives

The primary aim of this project is to design and develop a user-friendly web-based Learning Management System (LMS) that facilitates effective course management for instructors and provides an engaging, interactive learning experience for students. The system is intended to bridge the communication gap between educators and learners, support independent learning, and offer features that enable tracking of academic progress and performance in a structured, intuitive environment.

To achieve the stated aim, the project is guided by the following specific objectives:

- The system will provide a dedicated interface for instructors to design and organize course content, including text materials, video lectures, and supporting documents. It will also include functionalities to edit and delete existing content, enabling flexible and efficient course delivery.
- The LMS will include a personalized dashboard where students can monitor their course completion status, view upcoming tasks, and revisit learning materials. Interactive elements such as discussion sections, comment threads, and question submissions will enhance engagement and support deeper understanding. The platform will incorporate a quiz module that allows instructors to create multiple-choice, short-answer, or other question formats. Students will receive immediate feedback upon submission, helping them identify areas of improvement while reducing the instructor's manual grading workload.

1.4 Scope and Limitaiton

This system is designed as a **web-based e-learning platform** tailored for three primary user roles: **Administrators, Teachers, and Students**. The main purpose of the system is to facilitate online course creation, enrollment, learning, and performance tracking.

- **Admin Funtionalities**

Administrators have full control over the management of users and course data. Their capabilities include:

- **Create Teacher Account:** Admins can register and manage teacher accounts with login credentials and user roles.
- **Create Student Account:** Admins can register and manage student accounts, including editing or deleting user details when necessary.
- **Manage Courses:** Admins can oversee all courses in the system, including activating, deactivating, or removing inappropriate or outdated content.
- **Visualize System Data:** Admins have access to dashboards and analytical tools that display data trends such as student enrollment rates, teacher activity, course popularity, and other metrics for monitoring system performance.

- **Student Funtionalities**

Students use the platform primarily for learning purposes. Their key functions include:

- **Login/Sign Up:** Students can register a new account or log in using their credentials.
- **Enroll in or Purchase Courses:** Students can browse available courses and either enroll in free ones or purchase paid content through the system.
- **Post Reviews and Ratings:** After completing a course, students can provide feedback by posting reviews and assigning ratings to help other users make informed decisions.
- **Learning Activities:** Students can access course materials, take quizzes, and track their own progress within the platform.

- **Teacher Funtionalities**

Teachers are responsible for content creation and tracking student performance. Their main responsibilities are:

- **Create Courses and Quizzes:** Teachers can develop course modules, upload materials (videos, PDFs, etc.), and create interactive quizzes to assess learners.
- **Track Student Progress:** Teachers can monitor each enrolled student's performance, quiz scores, course completion status, and engagement level.
- **Visualize Course Data:** Teachers can view analytics and reports related to their courses, such as student completion rates, average quiz scores, and course ratings.

Overall, the system provides a strong foundation for digital learning, with clear room for expansion as user needs and technology evolve.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Research Topic

This research focuses on the development and implementation of a web-based e-learning system designed to cater to three distinct user groups: administrators, teachers, and students. The system aims to provide a centralized platform that simplifies the processes of course creation, student enrollment, learning material access, and performance evaluation. With role-specific functionalities, the platform enhances operational efficiency and user experience for all stakeholders involved in the learning process.

2.1.1 Design and Usability

The e-learning system is designed with a user-centered approach to ensure a clean, intuitive, and accessible interface for administrators, teachers, and students. It supports easy navigation, task efficiency, and responsive design for various screen sizes. Key features include simplified admin controls, user-friendly tools for teachers, and streamlined student access. While it lacks a mobile app, the web interface works well on mobile browsers. The design emphasizes consistency, simplicity, and usability to enhance user satisfaction and system effectiveness.

2.1.2 Integration of Technologies

The system also incorporates open-source frameworks and libraries to streamline development and ensure scalability. Features such as automated grading, user analytics, and secure authentication are integrated to improve performance and user engagement. By leveraging responsive design and cloud-based hosting, the platform ensures accessibility, reliability, and scalability, making it suitable for a wide range of educational institutions.

2.2 Previous Studies and Findings

As part of this research, it is important to acknowledge that several established e-learning platforms already exist, offering similar goals and functionalities. Understanding their features and target users helps highlight the unique value and positioning of the proposed platform. Two widely recognized platforms are Moodle and Google Classroom:

- **Moodle** is a widely used open-source learning management system (LMS) primarily designed for universities, colleges, and other educational institutions. It is renowned for its high degree of customizability, allowing institutions to adapt the platform according to their specific academic requirements. Moodle supports a broad range of plugins and modules, enabling features such as advanced grading systems, activity tracking, flexible course structures, user role management, and detailed analytics.
- **Google Classroom** is an online learning platform primarily designed for schools, teachers, and students, especially at the primary and secondary education levels. As part of the Google Workspace for Education suite, it offers seamless integration with widely used tools such as Google Docs, Google Drive, and Gmail. This integration enables teachers to easily create, distribute, and grade assignments while facilitating smooth communication with students, all through a clean and user-friendly interface.

2.3 Theoretical Framework

In designing and developing a simple Learning Management System (LMS), educational theories and technology adoption models provide essential guidance. Three key learning theories constructivism, cognitivism, and social learning theory inform how the platform should support student engagement and effective learning. According to TAM, users are more likely to use a system if they find it easy to use and useful in improving their learning outcomes. Therefore, a simple LMS should prioritize user-friendly interfaces and demonstrate value through tools like progress tracking and personalized learning paths.

By applying these theories, the LMS can be developed not only as a functional system but also as an engaging and educationally effective platform.

CHAPTER 3

METHODOLOGY

3.1 Research Design

To develop the proposed web-based Learning Management System (LMS), the research design focuses on four key areas: data collection, development tools, algorithms and system architecture, and the technologies used.

a. Tools

The system development process incorporates a variety of modern tools to ensure an efficient workflow and robust implementation:

- **Visual Studio Code (VS Code):** Used as the primary integrated development environment (IDE), VS Code provides a powerful and flexible workspace for writing, editing, and debugging code. Its support for numerous extensions enhances productivity and code quality during development.
- **Postman:** A tool designed for testing and validating APIs, Postman enables developers to simulate requests and inspect responses, ensuring seamless communication between the frontend and backend components of the system.
- **Notion:** Employed for project planning and documentation, Notion supports task tracking, scheduling, and collaborative note-taking. It serves as a central workspace for organizing development progress and maintaining clarity throughout the project lifecycle.
- **Render:** This platform is used for deploying the application. Render offers an accessible and scalable hosting solution, simplifying the deployment process and ensuring that the application is reliably accessible to users.
- **GitHub:** A version control and code repository platform, GitHub facilitates source code management, collaboration among developers, and project tracking. It enables efficient coordination of development tasks and ensures a secure, backed-up codebase.

b. Development Technologies

The development of this web-based application employs a modern and scalable technology stack designed to provide a responsive, secure, and efficient user experience across all platforms.

- **Frontend Technologies**

The frontend is developed using:

- **React:** A widely-used JavaScript library for building user interfaces, offering a component-based architecture and efficient rendering.
- **TypeScript:** Enhances JavaScript by adding static typing, which helps reduce bugs and improves code maintainability.
- **Material UI (MUI):** A comprehensive UI component library that ensures a clean, responsive, and user-friendly interface design.

- **Backend Technologies**

The backend stack includes:

- **NestJS:** A progressive Node.js framework that provides a robust structure for building scalable and maintainable server-side applications.
- **GraphQL:** Used as the API query language, allowing efficient data retrieval and flexible interactions between the client and server.
- **PostgreSQL:** A powerful and reliable relational database system chosen for its strong performance and data integrity features.

- **Supporting Services and Tools**

To enhance functionality and ensure smooth system operations, several third-party services and tools are integrated:

- **JWT (JSON Web Token):** Implements secure user authentication and session management.
- **Stripe:** Enables secure and seamless payment processing for transactions within the system.
- **Cloudinary:** Used for video storage and management, providing optimized delivery and media transformation.
- **Firebase:** Serves as the storage solution for handling user-uploaded files and other documents.

c. Project Management

An **Agile development** model is followed to allow for flexibility and iterative development. This method supports ongoing improvements, modular deployments, and frequent feedback integration, enabling continuous enhancement of system features.



The LMS was developed using an Agile approach, with iterative sprints focusing on feature delivery and feedback integration.

Figure 1 Project Management

d. Server Architecture

The system uses a client-server architecture. The **frontend** is built with **ReactJS** and styled using **MUI**, and it's deployed on **Render** for public access. The **backend** is developed with **NestJS**, connected to a **PostgreSQL** database, and also deployed on **Render**. It uses **JWT** for user authentication and communicates via **GraphQL** APIs. Additionally, the backend integrates with third-party services: **Cloudinary** for video storage, **Firebase** for file storage, and **Stripe** for payment processing. This architecture ensures modularity, scalability, and smooth integration of essential services.

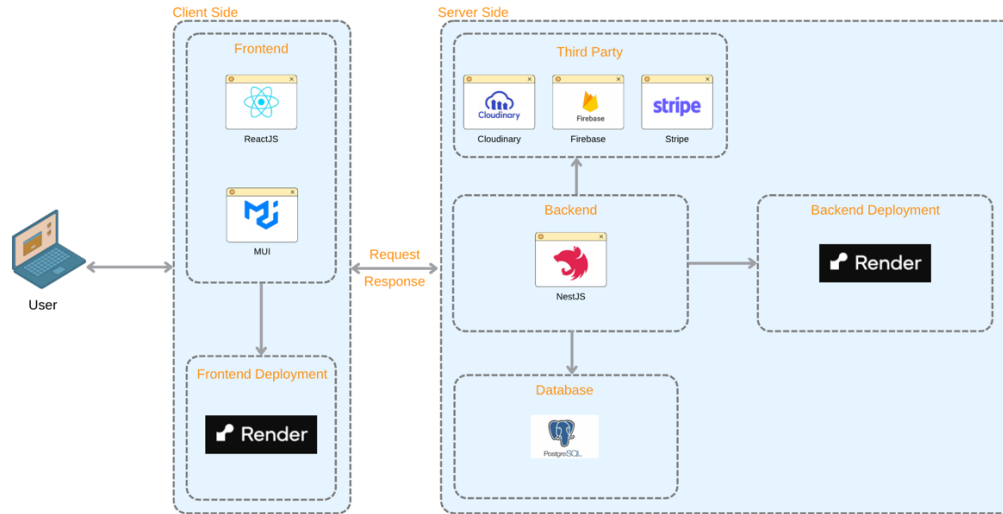


Figure 2 Server Architecture

e. Use Case Diagram

The diagram represents an online course platform where users can register, browse, and interact with courses. There are three main user roles: **admin**, **instructor**, and **general user**. After registration, users are identified by their roles. Admins can manage courses (create, edit, delete), instructors can create and manage their own courses, and general users can search, preview, enroll, watch videos, give ratings, and write reviews. The system supports both free and paid enrollment, with payment processing handled by an external **Credit Payment Service**. Users can also manage their profiles, and all activities are authenticated and validated through external services like **Authentication** and **Identity Provider**.

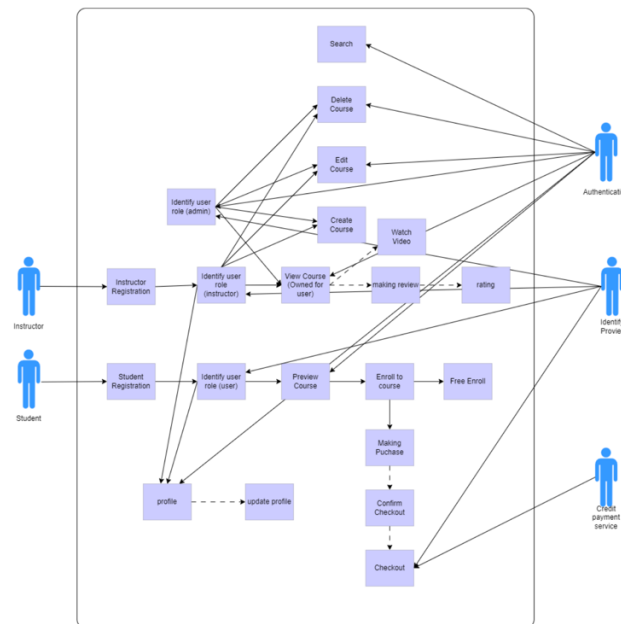


Figure 3 Use Case Diagram

3.2 Fuctional and Non-Functional Requirements

3.2.1 Functional Requirements

These are the essential functions that the system must provide to meet user needs. They represent the required operations, user interactions, and expected outputs that form the foundation of the platform's capabilities.

- Signup and Login (Authentication for Student and Teacher)
- Dashboard page for users
- Dashboard page for admin
- Browse page with course listings
- Search courses by title or keyword
- Filter courses by category
- View course details (videos, documents, reviews, rating, quiz)
- Enroll in courses (free or via VISA payment)
- Update user profile information
- Update username
- Update profile picture
- Upload profile picture
- Teacher add courses
- Teacher update courses
- Teacher delete courses
- Teacher add videos
- Teacher update videos
- Teacher delete videos
- Forgot password
- Logout

3.2.2 Non-Functional Requirements

The non-functional requirements of the platform focus on ensuring quality, performance, and reliability. The system must offer fast response times and handle multiple users efficiently, supporting scalability as user demand grows. Security is a key concern, with user data and transactions protected through secure authentication and encryption. The platform should also provide a user-friendly interface that is compatible across different devices and browsers. High availability, maintainability through modular design, and data integrity are also essential to ensure a smooth and dependable user experience over time.

3.3 Database Design

3.3.1 Schema design

The schema design is a fundamental component of the Learning Management System, as it defines how data is structured, stored, and related within the platform's database. A well-organized schema ensures the system can efficiently manage user information, course content, interactions, and transactions while supporting a smooth and responsive user experience. The LMS is built upon a relational database model that includes a set of interconnected tables. These tables represent core entities such as users, courses, enrollments, learning materials, and payments. Each table plays a specific role in maintaining data integrity and supporting the platform's main functionalities. For example, user data is managed and distinguished by roles such as administrators, teachers, and students; course data includes titles, descriptions, and learning materials; and enrollment data helps track student progress and participation in courses.

The schema is designed to support scalability and flexibility, allowing new features or modules to be integrated in the future without major structural changes. The relationships between tables such as one-to-many and many-to-many connections are implemented to ensure efficient data flow and interaction among users and content. Foreign keys and constraints are used to enforce data consistency and avoid duplication or conflict.

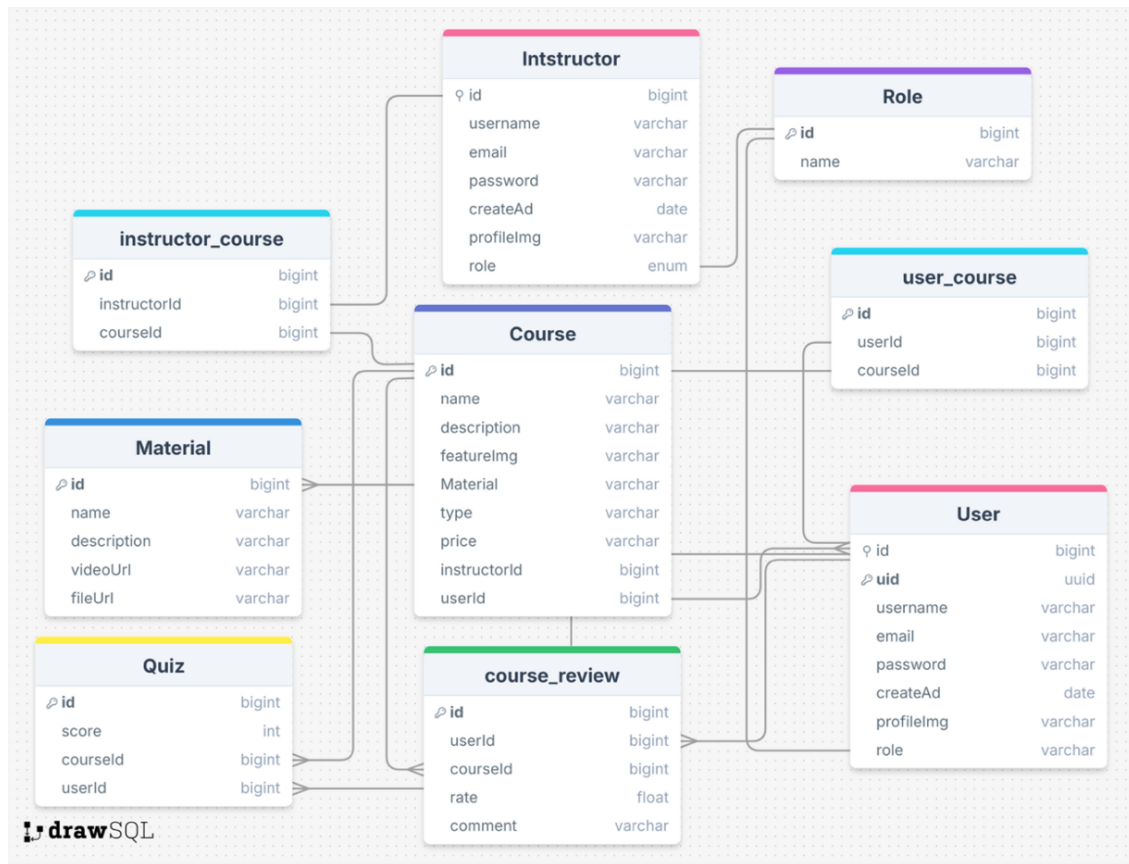


Figure 4 Database Relation

3.3.2 Data Relationship

The data relationships within the Learning Management System are designed to reflect real-world interactions between users, learning materials, and system functionalities. These relationships ensure that data is not only well-structured but also interconnected in a way that enables efficient operations, accurate reporting, and meaningful user experiences. The system is built around three main user roles **Administrators, Teachers, and Students** each of which interacts with the system in different ways. All user information is stored in a central user table, which includes key identifiers such as username, email, and role. This unified user structure simplifies role management and enables consistent authentication and authorization across the platform.

The system also supports interactive features through **user-generated content**, such as comments and reviews. Students can provide feedback, ask questions, or participate in discussions within a course or video page. These interactions are linked back to the individual user accounts, allowing instructors and admins to monitor activity and respond accordingly. Furthermore, each comment can have **nested replies**, which supports dynamic conversations and peer-to-peer support. These relationships are reinforced with foreign key constraints to maintain data integrity and enable complex queries for analytics, performance tracking, and user behavior insights. For example, teachers can view how many students are enrolled in a course, how far they've progressed, and how they've performed in quizzes. Similarly, students can view their learning history, course ratings, and personal progress dashboards.

Overall, the thoughtful design of data relationships ensures that the system is not only functional but also scalable and user-friendly. By mirroring real-world educational interactions, the platform can deliver a seamless and connected experience for all users involved.

CHAPTER 4

RESULT

4.1 Overview of the System Result

Each role is designed with specific capabilities to ensure a smooth and effective user experience. Admins oversee system operations and user management, teachers are empowered to create and manage educational content, and students can access courses and track their learning progress.

4.1.1 Admin Role

The Admin Dashboard in the Learning Management System offers a centralized overview of key statistics, including the number of students, instructors, courses, and categories, along with interactive charts for role distribution and activity trends. It allows administrators to monitor system performance and navigate to key management areas. The Admin All Courses section provides a detailed list of available courses with tools to add, edit, or remove them, while the List of Teachers page enables efficient management of instructor profiles, including their contact information and assigned courses. Together, these features streamline system administration and ensure smooth course and user management.

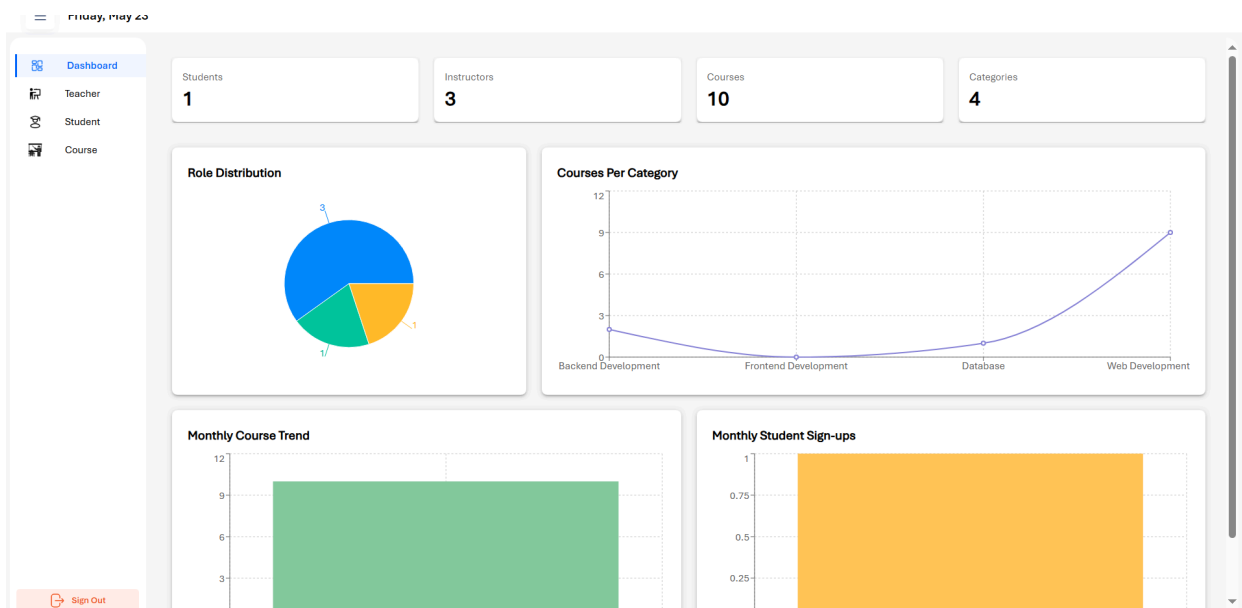


Figure 5 Admin Dashboard

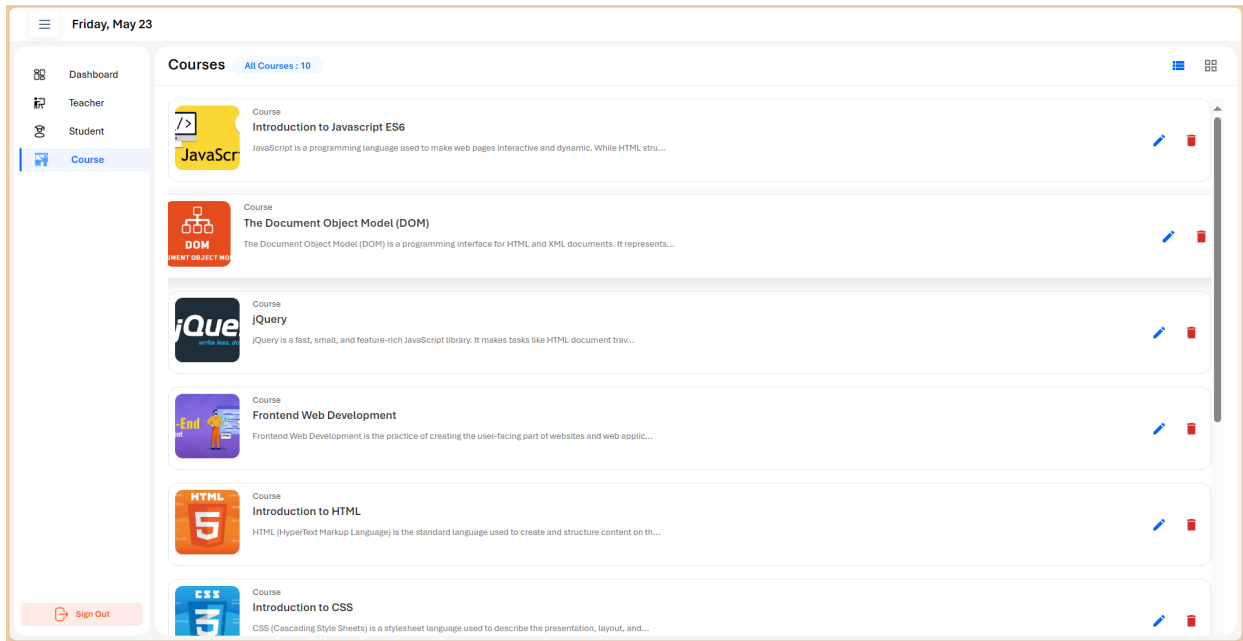


Figure 6 All Courses

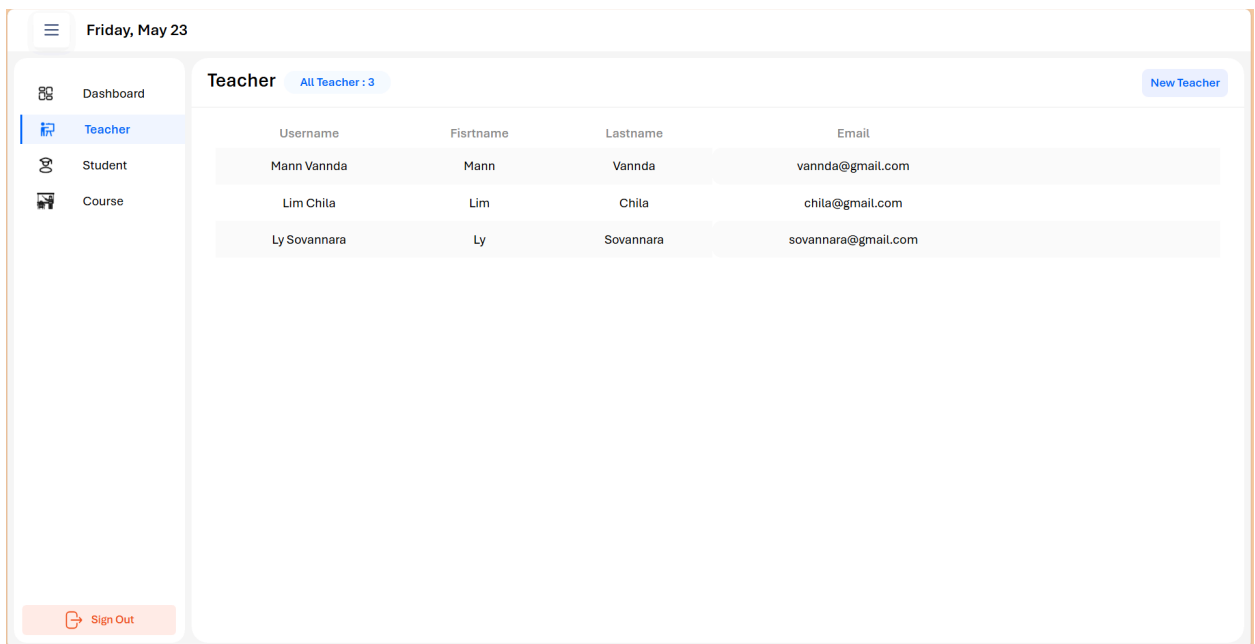


Figure 7 List of Teacher

4.1.2 Teacher Role

The Teacher Role in the Learning Management System is designed to empower instructors with tools to effectively manage and deliver educational content. Teachers have access to a **Course Creation** feature, allowing them to create new courses by adding titles, descriptions, categories, and learning materials such as videos, documents, and quizzes. Once courses are published, instructors can use the **Course Tracking** functionality to monitor student engagement and progress, including completion rates, quiz performance, and participation levels. These tools help teachers tailor their instruction, identify struggling learners, and enhance the overall learning experience.

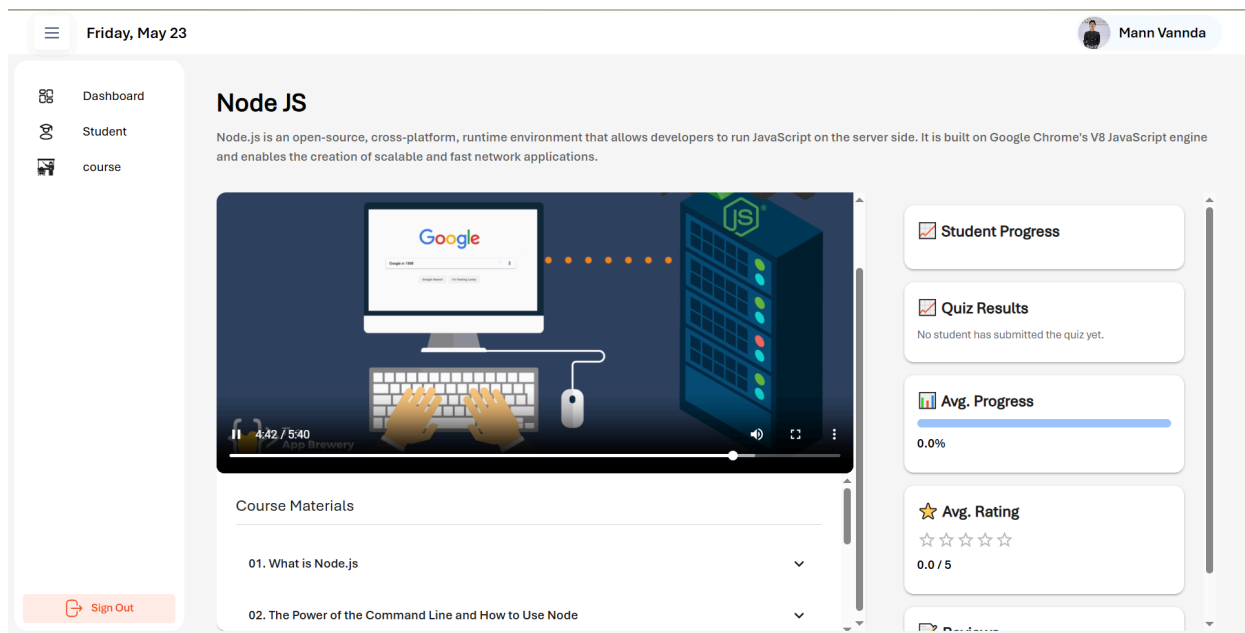


Figure 8 Course Tracking

Friday, May 30

Mann Vannda

New Course

1 Course Information 2 Course Material 3 Document 4 Quiz 5 Pricing

Course Information

Course Name
Enter course name

Category
Please Select

Level
Please Select

Description
Type course description...

Cover Image

No cover image selected

Upload Cover

Back Next

Sign Out

Figure 9 Create Course

4.1.3 Student Role

The Student Role in the Learning Management System is tailored to support learners in accessing and engaging with educational content. Upon logging in, students are directed to their **Student Homepage**, which provides a personalized dashboard showing enrolled courses, progress status, and upcoming activities. Through the **Student Material** section, learners can access course content such as lecture videos, documents, and assignments at any time. Additionally, the **Student Enroll with Payment** feature enables students to browse available courses, view details, and securely complete enrollment by making payments through the integrated payment system. This role ensures a smooth, self-paced learning journey with easy access to materials and streamlined course registration.

Friday, May 23

Lim Bunhov

My Learning

Courses

Favorite

Good Night, Lim Bunhov

welcome to Trenning, Check your priority learning

In Progress Learning Contant

See All

JavaScript

Course

Introduction to Javascript ES6

☆☆☆☆

Material

10

Completion

0%

Start

DOM

Course

The Document Object Model (DOM)

☆☆☆☆

Material

6

Completion

0%

Start

2 Learning Content

120 Learning Content

120 Learning Content

Recommend For You

See All

SQL

Introduction to SQL

SQL (Structured Query Language) is the standard

node.js

Build Your Own RESTful API From Scratch

A REST API (Representational State Transfer Application)

node.js

Node JS

Node.js is an open-source, cross-platform, runtime

Figure 10 Student Homepage

Friday, May 23

Lim Bunhov

Back

Enrolled

Project

DOM

index.html

index.js

styles.css

index.html

```

1 <!DOCTYPE html>
2 <html lang="en" dir="ltr">
3   <head>
4     <meta charset="utf-8">
5     <title>My Website</title>
6     <link rel="stylesheet" href="styles.css">
7   </head>
8   <body>
9
10    <h1 id="title">Hello</h1>
11
12    <input type="checkbox">
13
14    <button class="btn">Click Me</button>
15
16    <ul id="list">
17      <li class="item">a href="https://www.google.com">Google</a>
18      <li class="item">second</li>
19      <li class="item">third</li>
20    </ul>
21
22    </body>
23  </html>
24
25  <script src="index.js" charset="utf-8"></script>
26
27  </script>
28  </html>
29

```

Materials

6 lectures • 41m

01. Adding Javascript to Websites

10:44

0% watched

02. Selecting HTML Elements with Javascript

14:17

0% watched

03. Manipulating and Changing Styles of HTML Elements with Javascript

05:01

0% watched

04. The Separation of Concerns_ Structure vs Style vs Behaviour

05:57

0% watched

The Document Object Model (DOM)

The Document Object Model (DOM) is a programming interface for HTML and XML documents. It represents the structure of a web page as a tree of objects that can be accessed and manipulated using programming languages like JavaScript.

Figure 11 Student Material

20

← New business sandbox [Sandbox](#)

Choose a currency:

 KHR 41,590.17

 **\$9.99**

Exchange rate and fees of your bank may apply

Course #de79d07c-963c-4eaf-ab58-ede6524182d7 **\$9.99**

Pay with card

Email

email@example.com

Card information

1234 1234 1234 1234



MM / YY

CVC



Cardholder name

Full name on card

Country or region

Cambodia



☐ Securely save my information for 1-click checkout

Pay faster on New business sandbox and everywhere
Link is accepted.

Pay

Powered by **stripe**

[Terms](#) [Privacy](#)

Figure 12 Student Enroll with Payment

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 Discussion

The Learning Management System developed in this project serves as a practical demonstration of how modern web technologies can be applied to solve real-world problems in the educational sector, especially for small to medium-sized institutions with limited resources. The system was intentionally designed with a focus on simplicity, usability, and accessibility to bridge the technological gap often faced by educators and learners in such environments. From the early stages of development, the project was guided by a clear understanding of the existing limitations in widely-used platforms. Traditional LMSs like Moodle, although feature-rich and flexible, often come with steep learning curves, complex configurations, and higher technical requirements, which can be a barrier for institutions without dedicated IT support. On the other hand, platforms such as Google Classroom are more user-friendly but lack advanced functionalities such as real-time progress analytics, modular content delivery, or built-in payment integration. Recognizing this gap, our system aimed to find a middle ground delivering essential LMS features in a lightweight and easy-to-deploy package. A comparative review of existing systems reaffirmed the value of our tailored approach. While enterprise-grade LMS platforms offer extensive features, they are not always cost-effective or practical for institutions operating in developing regions. Our system, by contrast, presents a balanced solution with core functionalities like course management, content delivery, interactive assessments, and progress tracking—while remaining simple, maintainable, and affordable.

In summary, this project not only demonstrated the effective use of modern web technologies in educational software development but also emphasized the importance of designing for context. Through iterative design, thoughtful technology selection, and user-centered development, we successfully delivered a system that addresses common pain points in online education and offers a viable alternative for institutions seeking digital transformation without overwhelming complexity.

5.2 Conclusion

In conclusion, the development of this Learning Management System (LMS) marks a significant step toward providing an accessible, efficient, and sustainable digital learning solution tailored to the unique needs of small to medium-sized educational institutions. Throughout this project, our primary objective was to bridge the gap between the growing demand for online education and the limited availability of affordable, user-friendly platforms—especially in regions with constrained technological resources. The LMS we developed focuses on core functionalities that directly support the teaching and learning process, such as course creation, content delivery, student enrollment, progress tracking, and assessment management. These features are integrated into a streamlined and intuitive interface that ensures ease of use for students, teachers, and administrators alike. By deliberately avoiding unnecessary complexity and focusing on real-world usability, the system is designed to be deployed and maintained with minimal technical overhead, making it highly suitable for institutions without specialized IT departments. We believe that this platform not only addresses immediate academic challenges but also supports long-term educational development. It encourages active learning, self-paced study, and real-time feedback, all of which are essential in nurturing independent, motivated learners. It also empowers educators to monitor student engagement and adjust their instructional strategies more effectively. Unlike more complex or expensive systems, this LMS provides a lightweight alternative that promotes digital education in a manageable and inclusive way. As digital learning becomes increasingly important in today's educational landscape, especially in developing regions, this system will empower institutions and learners to embrace modern technology with greater ease. Through this initiative, we aim to support educational growth, reduce digital barriers, and contribute to a more equitable and connected learning environment for all. Ultimately, this project has shown that meaningful technological solutions do not need to be overly complex or costly to have a real impact. Through thoughtful design, collaborative development, and a focus on user needs, we have created a system that is practical, scalable, and aligned with the goals of modern education. We hope that this LMS will serve as a foundation for future innovation, contributing to a more connected, empowered, and digitally-enabled learning environment for all.

5.3 Future Work

As part of our commitment to continuous improvement and adaptability, several future enhancements have been identified to further strengthen the functionality, user experience, and security of our Learning Management System (LMS). These planned developments aim not only to refine the technical performance of the platform but also to enrich its educational value and accessibility for all users. Another planned feature is the integration of a course-specific comment and discussion section. This enhancement is designed to promote interaction and engagement between students and instructors. By allowing users to post questions, share insights, and offer feedback directly within each course page, the platform will support a more dynamic and collaborative learning environment. Such forums encourage peer-to-peer learning, increase student involvement, and enable instructors to address issues or clarify content in real time. This feature also aligns with social learning theories, which emphasize the value of community and interaction in the educational process. In addition to new features, we recognize the need for ongoing efforts to enhance the user interface (UI) and overall user experience (UX). While the current design emphasizes simplicity and clarity, future updates will focus on making the interface more visually appealing, intuitively navigable, and responsive across various devices and screen sizes. Planned improvements include a redesigned navigation menu, better categorization of content, interactive visual elements, and accessibility considerations such as larger buttons, proper color contrast, and keyboard navigability. These enhancements are intended to make the platform more inclusive and usable for individuals with different needs and technical skill levels. Finally, we also plan to conduct user testing and feedback collection after deployment to guide future development. By involving end-users in the improvement process, we ensure that changes are grounded in real-world needs and preferences, leading to a more responsive and user-centered platform.

In summary, the future work outlined here reflects our long-term vision for a secure, interactive, and accessible LMS. By focusing on enhancements that support engagement, usability, and scalability, we aim to ensure that the system continues to evolve in line with educational trends, user expectations, and institutional goals.

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